

ICS 4111: Embedded Systems & IoT

Practical Exercise 2 — Part A: Circuit Calculations

Circuit A: Series Circuit

Supply Voltage: 125 V | Resistors: $R_1 = 20\ \Omega$, $R_2 = 30\ \Omega$, $R_3 = 50\ \Omega$

1. Total Resistance

In a series circuit, resistances add directly:

$$R_{\text{total}} = R_1 + R_2 + R_3 = 20 + 30 + 50 = 100\ \Omega$$

2. Total Current from Power Supply

Using Ohm's Law ($I = V / R$):

$$I_{\text{total}} = 125\ \text{V} / 100\ \Omega = 1.25\ \text{A}$$

3, 4 & 5. Current, Voltage Drop, and Power per Resistor

In series, the same current (1.25 A) flows through every resistor. Voltage drop = $I \times R$, Power = $V \times I$.

Resistor	Voltage Drop (V)	Current (A)	Power (W)
$R_1 = 20\ \Omega$	25.00	1.25	31.25
$R_2 = 30\ \Omega$	37.50	1.25	46.88
$R_3 = 50\ \Omega$	62.50	1.25	78.13
Total	125.00	1.25	156.25

Circuit B: Parallel Circuit

Supply Voltage: 125 V | Resistors: $R_1 = 20\ \Omega$, $R_2 = 100\ \Omega$, $R_3 = 50\ \Omega$

1. Total Resistance

For parallel resistors, use the reciprocal formula:

$$1/R_{\text{total}} = 1/20 + 1/100 + 1/50 = 5/100 + 1/100 + 2/100 = 8/100$$

$$R_{\text{total}} = 100/8 = 12.5\ \Omega$$

2. Total Current from Power Supply

$$I_{\text{total}} = V / R_{\text{total}} = 125 / 12.5 = 10 \text{ A}$$

(This also equals the sum of all branch currents: $6.25 + 1.25 + 2.5 = 10 \text{ A}$ ✓)

3, 4 & 5. Current, Voltage Drop, and Power per Resistor

In parallel, every resistor sees the full supply voltage (125 V). Branch current = V / R .

Resistor	Voltage Drop (V)	Current (A)	Power (W)
R1 = 20 Ω	125.00	6.25	781.25
R2 = 100 Ω	125.00	1.25	156.25
R3 = 50 Ω	125.00	2.50	312.50
Total	125.00	10.00	1250.00

Note: Power values are high because of the 125 V supply — particularly R1 (20 Ω) which draws 6.25 A in the parallel circuit.

Part B

BOM Diagrams

(a)

Designator	Name	Value	Quantity
V1	Voltage Source	125V	1
R1	Resistor	20 Ω	1
R2	Resistor	30 Ω	1
R3	Resistor	50 Ω	1

(b)

Designator	Name	Value	Quantity
V1	Voltage Source	125V	1
R4	Resistor	20 Ω	1
R5	Resistor	30 Ω	1
R6	Resistor	50 Ω	1

(2)

Designator	Name	Value	Quantity
U1	Arduino Nano	Arduino Nano	1
U2	DHT22	DHT22	1
R7	Resistor	10k Ω	1